

Flag Challenge — Two-Tier Structure (Free / \$1 Upgrade)

Free Tier — No Account, No Data Leaves the Device

- Local score storage only (best streak, accuracy, last session)
- No login screen — just play
- Satisfies COPPA trivially since nothing personal is collected

Paid Tier (\$1 one-time) — Unlocks Cloud Features

- Purchase triggers account creation (email or just a username + device-linked ID)
- Score history synced across devices
- Global/friend leaderboards
- The \$1 purchase acts as a soft age/intent filter — requiring a payment method screens out most under-13 users without a formal age gate for this tier

User Flow

1. App opens straight into gameplay — no forced signup
2. After a session, show local score with a subtle prompt: "Save your history + join the leaderboard — \$1, one time"
3. On purchase: create account (simplest = anonymous UUID + optional email for recovery, not required)
4. Sync local scores retroactively into the cloud record so prior progress isn't lost

Technical Shape

- **Local:** on-device storage (UserDefaults / SharedPreferences or a simple JSON file) for free-tier scores
- **Cloud:** lightweight backend — Supabase is a good fit:
 - One `users` table
 - One `scores` table
 - Row-level security so users only read/write their own rows
 - A `leaderboard` view
- **Auth:** Supabase Auth — anonymous auth by default, upgrade to email on purchase if the user

wants recovery

- **IAP:** App Store / Play Store in-app purchase API confirms payment, then flips a local flag that unlocks the sync/login UI

Privacy Posture

- No ad SDKs on either tier
- Paid tier's account data (username + scores) never leaves the Supabase instance — no third-party sharing
- Suggested one-line privacy policy note:

"Free play requires no account. Paid users' data is used only for score sync and leaderboards, never shared or sold."

Open Design Decision

Should leaderboard entries use a real display name/username, or auto-generated anonymous handles (e.g. "SwiftFalcon42")?

Anonymous handles avoid name-based moderation/privacy headaches and are simpler to launch with.

Next Step (when ready)

Supabase schema (users / scores / leaderboard tables + RLS policies), ready to wire up.